

2-Mark Questions & Answers — Data Structures Module 3: Linked Lists

Q1. What is a Linked List? State its node structure.

A **Linked List** is a dynamic linear data structure composed of nodes allocated in non-contiguous memory, where each node contains **Data** and a **Pointer** (**next**) storing the address of the subsequent node.

Q2. Differentiate between Array and Linked List.

Feature	Array	Linked List
Memory Allocation	Static / Contiguous memory.	Dynamic / Non-contiguous memory on heap.
Insertion / Deletion	Slow O(N) (requires shifting elements).	Fast O(1) (updating pointer links).
Element Access	Direct random access O(1) .	Sequential access O(N) .

Q3. What is a Doubly Linked List (DLL)?

A **Doubly Linked List** has nodes containing two pointers: **prev** (pointing to predecessor) and **next** (pointing to successor), allowing bidirectional traversal.

Q4. What is a Circular Linked List?

A **Circular Linked List** is a linked list where the **next** pointer of the last node points back to the **Head node** instead of **NULL**.

Q5. How is a Stack implemented using a Singly Linked List?

Pushing and popping elements are performed at the **Head of the Linked List** in **O(1)** time (**push** : insert at beginning, **pop** : delete from beginning).